

The empty chair: a system for measuring epistemic absence in open-commons retrieval

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Abstract

Retrieval-augmented and "deep research" AI systems answer contested questions by synthesizing whatever sources they can reach. They report what the literature says; they are silent about what it cannot say. This paper introduces TACET, a system that audits a contested question against the open-licensed (CC-BY) literature and produces not an answer but a structured map of what that literature can and cannot support. TACET's central instrument is the empty chair: a measurement of which expected perspectives are structurally absent from the open commons, taken against an expectation declared before retrieval. Two independent readers, neither assigned a side, score each claim; their convergence is reported as coherence, never as truth. Across four frozen case studies, the live-crux signal never fires: on open corpora the readers either converge or abstain, and a recurring pattern emerges — theses living in books and in non-English languages are systematically invisible to the open commons, a measurable structural bias that answer-giving systems hide. We frame this as an operational response, by measurement rather than curation, to the algorithmic epistemic violence diagnosed in prior work. All artifacts are frozen and reproducible.

Keywords: epistemic absence; open-commons bias; retrieval audit; coherence; abstention; algorithmic epistemic violence; reproducibility

1 Introduction

A researcher asks an AI system a contested question and receives a fluent synthesis of what the literature says. The synthesis reports presence: the sources it found, the position they support. It is silent about absence: what it could not find, and why. For a contested question, what the literature cannot support is as decision-relevant as what it can — yet retrieval-augmented and "deep research" systems draw no boundary around the knowable. Completeness is claimed by omission.

This paper presents TACET, a system that audits a contested question against the open-licensed (CC-BY) literature and returns not an answer but a structured map of what that literature can and cannot support. The system's central instrument is the empty chair: a measurement of which expected perspectives are structurally absent from the open commons, taken against an expectation declared before retrieval. Two independent readers, neither assigned a side, score each claim; the system reports their convergence as coherence, never as truth.

We make five contributions. First, TACET itself: a working system with frozen, reproducible artifacts. Second, the empty chair, an operational measurement of structural absence. Third, four frozen case studies that span the space of outcomes a coverage audit produces. Fourth, a two-layer comparison against a deep-research baseline that distinguishes reachability from verification

and makes hidden dependency legible. Fifth, an empirical finding: across the four cases, the open commons exhibits a measurable, structural bias against theses bound to books and to non-English languages.

This work is the constructive sequel to a diagnosis. Prior work (Bonomo & Donard, 2026) names algorithmic epistemic violence: the structural consequence of generative systems trained on corpora naturalized as neutral technical data, which reproduce at industrial scale the epistemic asymmetries that constituted them. That work proposes a curated corpus as one response — building a corpus that includes what the commons excludes. TACET takes a different path: rather than curate inclusion, it measures exclusion and makes it legible. The two are complementary responses to the same diagnosis.

2 Background and related work

The concept of epistemic violence originates with Spivak (1988/2010) and is radicalized as epistemicide by Santos (2014); we draw on their articulation as developed for the algorithmic setting by Bonomo and Donard (2026), rather than re-deriving it here. A growing literature names the harm in AI systems specifically: Kay, Kasirzadeh, and Mohamed (2024) propose generative algorithmic epistemic injustice; Mollema (2025) proposes generative hermeneutical erasure as an automation of epistemicide. A parallel tradition documents algorithmic bias empirically: Noble (2018) on search engines, Benjamin (2019) on how digital systems recode hierarchy under an appearance of neutrality.

On the technical side, retrieval-augmented generation (Lewis et al., 2020) grounds model output in retrieved sources, and attributed question answering (Bohnet et al., 2023) argues that the capacity to attribute generated text to verifiable sources returns to the user the agency to inspect a source and judge its reliability.

Prior work either names the harm or curates a corpus to include what the commons excludes. TACET does neither. It measures the exclusion and makes it legible without claiming to fill it. The empty chair is, methodologically, the inverse of the curated corpus: curation decides what will exist; the empty chair measures what the commons decided would not.

3 System design

TACET runs a contested question through a fixed pipeline. Each stage is constrained by a design commitment that we can state and defend.

Open-licensed only. TACET reads abstracts under a CC-BY license, redistributable with attribution. It does not reach paywalled papers or most books. We keep this limitation on purpose: confining the system to the open commons turns the corpus into an instrument that can measure what the commons omits (§6). A system that quietly reaches past the paywall gains completeness and loses the ability to tell the reader that the source convincing them is one they cannot themselves verify.

Pre-registered expectation. Before any retrieval, a reference hypothesis and an expected-coverage baseline are written down and accepted by a human. The reference hypothesis has two clauses: the best-sustained position and the structural concession that keeps the question open. The coverage audit measures the gathered corpus against this pre-declared expectation, never the reverse. This anti-circularity is what lets an empty chair be a finding rather than an artifact of what happened to arrive. The harvest cap — how many open sources are gathered — is part of this pre-registration: declared at step 0 and frozen with the protocol, never tuned after seeing the result. Running the same question under different frozen caps separates a structural absence, where the

empty chair persists from fifty sources to two thousand, from an absence of cut, where it appears only because the cap was low.

Two undecided readers. The same corpus is read twice, by readers from distinct model families. Each scores each claim as supports, contradicts, or insufficient, relative to the reference hypothesis. Neither reader is assigned a side. The system compares the two and labels each claim: robust core when both lean the same non-insufficient way, live crux when they lean opposite ways, unsupported when at least one abstains. The readers’ verdicts are juxtaposed, never fused into a single voice — divergence is preserved as information, not averaged away.

The empty chair. The coverage audit compares the corpus against the pre-declared expectation and reports, per expected dimension, one of three states: present, measured-absent (an empty chair: searched for and not there), or not-measured (no basis to assess, as when a language boundary blocks the lexical gate). The distinction matters: the system never reports a not-measured dimension as an empty chair, and never reports an empty chair as mere absence of data. It locates structural silence against a declared rule.

Coerced narrative. When TACET emits prose, a deterministic verifier binds that prose to the map. A numeric guard checks that every figure in the narrative traces to the convergence and coverage data; a thematic sieve blocks claims the structure does not support. The narrative cannot assert what the map does not carry.

Reproducibility. Every case is frozen as a fixture and replayed offline and deterministically; network access occurs only in the preparation steps, never in replay. An executable auditor re-checks the frozen numbers against the rules that produced them — a number without its rule attached is treated as a defect. The auditor is itself auditable: on its first run it flagged imprecisions in the authors’ own fixtures and refuses to score a fixture whose schema it does not recognize.

4 Core concepts

TACET’s contribution is a set of concepts made operational — each is a measured mechanism in the pipeline, not a claim about the world.

The empty chair. Epistemic absence as a measurement. The empty chair is the count and identity of expected perspectives that the open corpus does not contain, taken against an expectation declared before retrieval. It is the inverse of a curated corpus: where curation decides what will exist, the empty chair measures what the commons decided would not.

Coherence, not truth. The system reports whether two independent doubts, reading the same evidence, reach the same place. It never reports that the place is correct. Two readers converging tells you their doubts moved alike; it does not tell you they were right. Where the two readers share a bias, their convergence can be convergence-by-shared-bias — a limit we name in §7, not one we claim to have closed.

Open-commons bias. The empirical finding of §6, named as a concept: the open commons is not a neutral sample of what is known. It systematically excludes theses bound to books and to non-English languages, and TACET measures that exclusion rather than reproducing it as neutral availability.

Two undecided readers. Convergence from independent doubt. The system does not bet on a single reader being neutral — machine neutrality is impossible. It bets on two readers from distinct families being decorrelated, measures their divergence rather than averaging it, and uses the empty chair to locate the bias that lives in the commons rather than in the readers.

Abstention over fabrication. At four distinct points the engine declines to manufacture knowledge: the readers abstain per claim; the coverage audit separates measured absence from the

unmeasurable; the relevance gate abstains when it has no basis to judge; and the narrative refuses to assert beyond the map. The system is exact about the boundary of what it knows.

5 Case studies

We evaluate TACET on four contested questions, each frozen as a fixture: the reference hypothesis, the harvested corpus, both readers’ per-claim verdicts, the convergence map, and the coverage audit are stored and replayed offline and deterministically. Every number below is read from a frozen fixture; none is recomputed at read time. The four cases were chosen to span the space of outcomes a coverage audit can produce — base present, base absent, dispute preserved, and thesis invisible — rather than to showcase a single behavior.

The two readers are drawn from distinct model families: reader A is ‘nvidia/nemotron-3-nano-30b-a3b’, reader B is ‘openai/gpt-oss-120b’. Neither is assigned a side. Each scores each claim as *supports*, *contradicts*, or *insufficient* relative to the pre-declared reference hypothesis. A claim is *robust core* when both readers lean the same non-insufficient way, *live crux* when they lean opposite ways, and *unsupported* when at least one abstains.

COVID-19 origin (base present). Reference hypothesis: natural zoonotic spillover is the best-currently-supported reading, with the question inconclusive and a research-related accident neither proved nor excluded. Over a 44-claim corpus, the two readers converge to 8 robust-core, 0 live-crux, and 36 unsupported. The convergence was not engineered: two readers from different companies, with no assigned position, independently reached the same lean on the phylogenetic evidence. This is the anchor case — it shows that the method is not only abstention; when the open evidence carries a reading, the system registers it. The relevance gate returns *not-assessed* — the lexical gate is English-only by design, a documented limitation, not a verdict on the corpus — with an aligned fraction of 0.068 because the corpus is Portuguese; the convergence is nonetheless measured on the evidence itself. Even here, in the easy case, the coverage audit records what is missing rather than asserting completeness.

LHC safety (base absent). Reference hypothesis: well-tested physical laws and astronomical observations imply that any TeV-scale black holes produced at the LHC would not threaten Earth. The corpus has 30 claims — 28 CC-BY abstracts plus two safety preprints ingested as non-redistributable (full text redacted; DOI and sha256 retained). The readers reach 1 robust-core, 0 live-crux, 29 unsupported, with the relevance gate aligned (0.967). They abstain on all 28 open-corpus claims; the lone robust-core both readers affirm is itself one of the ingested out-of-CC-BY preprints, the Giddings–Mangano analysis (arXiv 0806.3381). The abstention over the open commons is the finding, and it separates two distinct absences the open literature hides. The consolidated safety report (LSAG) is absent entirely — a measured empty chair of genre *report*. The analysis the conclusion actually leans on was readable only as a redacted, non-redistributable claim: present in provenance, but unverifiable by the reader. The deep-research baseline rested its reassurance on exactly those out-of-CC-BY preprints (arXiv 0806.3381, 0808.1415), alongside a blog, a YouTube video, and a Wikipedia article, without signaling that its foundation cannot be verified. TACET does not know more than the baseline; it is honest about what is not in the base.

Eggs and cardiovascular risk (dispute preserved). Over a 29-claim corpus, the readers’ verdicts compose to 2 robust-core, 0 live-crux, 27 unsupported; the gate is *not-assessed* (0.138, Portuguese corpus). The aggregate signal in the general population is null, and the live dispute — subgroups, long-duration trials — falls into the empty chair rather than being resolved. The decisive axis was verifiability: every TACET claim traces to a resolvable DOI (1.00), while the deep-research baseline produced fluent prose citing zero resolvable sources (0.00). The same question, audited two

ways: one side can be inspected, the other cannot.

Freud and the analytic session (thesis invisible to the open commons). This case is run twice. In the derived-query run, the relevance gate is *mixed* (0.476): the query drifted by homonymy toward Marxian surplus-value, and the readers abstain across all 21 claims (0 robust-core). In the focused-query run, the query is corrected and the gate reports aligned at 0.8 — lexically on-topic — yet the readers still abstain on 29 of 30 claims, and the single robust-core is spurious: a paper titled *Repair cafés in the Netherlands: Capitalist abstinence as a challenge to a linear capitalist economy*, matched on shared vocabulary, not on Freud. The actual thesis (developed by Tomšič, in Portuguese and German, in book form) sits in a triple empty chair: Portuguese, German, and book. Lexical alignment is necessary but not sufficient; a ruler can be on-topic and the thesis can still live in a book or a language outside the indexed perimeter, and only the empty chair reveals it.

The thread across the four cases. One result was not predicted by the design mock: the live-crux signal fired zero times across all four real corpora. On genuinely open corpora, the two readers either converge (robust core) or abstain (unsupported / empty chair); they have not yet diverged into a real crux. The intended center of TACET — two sides in measured conflict — is not what the open commons produces. What it produces instead is absence. This reframes the method’s central claim: the product is not an arbitrated dispute. The product is the empty chair.

6 Comparison with a deep-research baseline

The FLF case studies invite a comparison between assessment methodologies on the same questions, with explicit discussion of where they agree and diverge. We compare TACET against a deep-research baseline — a single-pass LLM synthesis with web access — on the four cases, holding the question fixed. The comparison is honest by construction: it measures fidelity of provenance, not which method reaches a better conclusion. No winner is declared on the conclusion; the deterministic axes report raw numbers and the judge axes are left to the reader. The baseline is a reproducible reference point, not a state-of-the-art ceiling.

Verifiability is reported in two layers. *Landing* asks whether each cited page is reachable; *registered* asks whether a cited DOI exists in the global DOI system, independent of publisher link rot. The asymmetry the comparison exposes is not in the headline fraction but in what the fraction means.

On eggs, every TACET claim resolves at both layers (landing 1.00, registered 1.00 across 29 CC-BY DOIs), while the deep-research output cited zero resolvable sources (landing 0.00, registered 0.00): fluent prose with no DOI and no extractable URL. On LHC, the numbers invert in a way that rewards reading the layers rather than the headline. The baseline scores landing 1.00 — all ten cited URLs exist — but those URLs include a blog, a YouTube video, and a Wikipedia article, and the output contains zero DOIs, so its registered layer merely mirrors landing rather than verifying anything. TACET scores landing 0.967 and registered 1.00: the single landing miss is link rot on a registered DOI (the publisher page returns 404 while the DOI still resolves at the handle), and all 30 academic identifiers verify at the registration layer. A headline "baseline 1.00 vs TACET 0.967" would invert the real finding. The baseline’s 1.00 is reachability of mixed-quality URLs without a verification layer; TACET’s 0.967/1.00 is link rot on a registered academic corpus. Reachability is not verification, and the two-layer report keeps them distinct.

The deeper divergence is in hidden dependency. On LHC the comparison records two out-of-CC-BY preprints the baseline rests its reassurance on, presented without signaling that this foundation cannot be verified by the reader; TACET marks the same preprints as redacted, non-redistributable claims and the consolidated safety report as an empty chair. This is the distinction

the FLF assessment frame asks for between what an investigation settles and what it merely performs settling. Attribution to verifiable sources is what returns to the reader the agency to inspect a source and judge it (Bohnet et al., 2023); a retrieval pass that grounds output in sources (Lewis et al., 2020) still inherits, silently, whatever its sources omit. The failure mode the baseline exhibits — fluent confidence over unverifiable or absent foundations — is the bibliographic hallucination and displaced attribution that the epistemic-injustice literature identifies in generative systems (Kay, Kasirzadeh & Mohamed, 2024; Mollema, 2025). TACET does not avoid this by knowing more. It avoids it by reporting the boundary of what its corpus can verify.

7 Finding: the open commons has a measurable, structural bias

Across the four cases, one pattern recurs: theses that live in books and in non-English languages are systematically invisible to the CC-BY commons. A scientific consensus can sit behind a paywall, as in the LHC safety argument; a humanities thesis can sit in a Portuguese-language book, as in the Freud case; and no query correction reaches either. The Freud focused run makes the point precisely: a lexically aligned query (0.8) still returns only a spurious convergence, because the thesis is not in the perimeter at all.

This is one specific, measurable face of what prior work names algorithmic epistemic violence (Bonomo & Donard, 2026): a corpus politics naturalized as neutral technical availability reproduces, as what is and is not retrievable, the asymmetries that constituted it. TACET does not correct this bias. By disciplining itself to the open commons and measuring the empty chair against a pre-declared expectation, it makes the bias legible rather than papering over it. Where a deep-research pass dissolves the gap into a fluent transition sentence, TACET names, counts, and anchors what the base did not cover.

For an investigator, this answers a question answer-giving systems hide: which questions can the open literature actually support, and which require reaching, deliberately and with provenance, beyond it. That capability is structural to the format. It does not shrink as base models improve — a better reader abstains more accurately on an absent source, it does not hallucinate the source into presence.

8 Compounding: the frozen artifact survives scrutiny

A structured analysis is worth more when it travels — when another investigator, possibly with different beliefs, can pick up where one left off, extend the corpus, and re-run the assessment without re-litigating the original work. TACET is built so that each case is a portable, inspectable artifact rather than a single-user trace.

Every case is frozen as a fixture: the reference hypothesis, the harvested corpus, both readers’ per-claim verdicts, the convergence map, and the coverage audit are stored together and replayed offline and deterministically. A separate investigator can take a frozen case, add sources to the corpus, and re-run the pipeline; because the expectation was declared before retrieval, their extension is measured against the same rule, and the empty chair either fills or persists as a comparable measurement. Multilingual retrieval is handled by a multi-language embedding model (Chen et al., 2025), so a corpus spanning Portuguese, French, English, Spanish, and German is indexed in one semantic space rather than siloed by language.

The artifact is also built to survive adversarial scrutiny, and the mechanism is itself part of the contribution. An executable auditor checks 56 invariants over the frozen fixtures: every reported number must trace to the rule that produced it, no claim may leak across the redaction boundary,

the verifiability layers must carry their structured notes, and the comparison fixtures must declare whether a registered fraction is a verification or a fallback. A number without its rule attached is treated as a defect. The auditor is itself auditable: on its first run it flagged imprecisions in the author’s own fixtures and a mis-scoped invariant the author had written, and it refuses to score a fixture whose schema it does not recognize. This is what lets the artifact compound rather than merely persist — a later user does not have to trust the numbers, they can re-derive them, and the auditor reports if any number and its rule have drifted apart.

9 Limitations

We name these rather than paper over them. The readers are not neutral, and TACET does not claim they are; machine neutrality is impossible, and claiming it would be the epistemic erasure the project investigates. The system bets on two readers from distinct families being decorrelated, measures their divergence, and uses the empty chair to locate the bias that lives in the commons. Where the two readers share a bias, their convergence can be scored as robust core when it is convergence-by-shared-bias — a named, unclosed limit; the defense is that distinct families and the observed sub-threshold divergence make full correlation unlikely, not impossible.

The relevance gate is lexical, a proxy for topicality that misses semantic drift sharing vocabulary; a semantic and multilingual gate is future work. The live-crux signal did not fire on any of the four real corpora, which is itself the central empirical result of §5 rather than a defect: on open corpora the readers converge or abstain, and the within-corpus divergence the design anticipated remains demonstrated only in unit tests. The crux is deliberately not resolved by having the two readers debate until one persuades the other: that would measure persuasion rather than evidence, collapse the independence the method depends on, and amplify shared bias into manufactured consensus. The honest enrichment of a crux is annotated divergence — each reader naming the evidence its reading anchored on, without moving the other — which is future work, kept distinct from any persuasion tool. Finally, the harvest cap shapes the corpus; it is declared and frozen at step 0, and varying it across runs is how the system separates structural absence from absence of cut.

10 Conclusion

TACET measures what it cannot support. Given a contested question, it audits the open-licensed literature, reports where two independent readers converge or abstain, and — through the empty chair — names which expected perspectives the open commons does not contain. It certifies coherence, never truth, and it prefers a named absence to a fabricated presence.

The four cases show the method spanning its outcome space: it registers a base when the evidence carries one (COVID), names the hidden dependency when the base sits outside the commons (LHC), preserves the dispute and anchors provenance where verifiability divides two methods (eggs), and measures the structural invisibility of a thesis bound to a book and a language (Freud). The thread across them is that the open commons produces absence more than dispute, and that absence is the product. The empty chair is a transferable instrument: any contested question audited against an open corpus can be asked not only what the literature says, but who is missing from the table.

Availability

The TACET implementation, the frozen case fixtures, the deep-research comparison fixtures, and the executable auditor are available in the project repository. Each case replays offline and deterministically from its fixture; the auditor re-checks the reported numbers against the rules that produced them. A curated run order (LHC, then eggs, then COVID, then Freud) reproduces the results reported here.

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Acknowledgments

The author thanks Véronique Donard for her active scientific supervision of this work and for the partnership in the broader research program from which it emerges. The concepts of Peripheral Listening and the *Periphérico* tradition are developed with Jairo Carioca de Oliveira and Ronald Lopes de Oliveira (Carioca, Lopes & Bonomo, in preparation).

Use of AI assistance

Conceptual development and drafting of this work were supported by an AI assistant used as a thinking tool, under the full direction and decision of the author. The system design, the concepts, and all claims are the author’s own.

References

- [1] Benjamin, R. (2019). *Race after technology: Abolitionist tools for the New Jim Code*. Polity Press.
- [2] Bohnet, B., et al. (2023). Attributed question answering: Evaluation and modeling for attributed large language models. arXiv:2212.08037.
- [3] Bonomo, H. A. R., & Donard, V. (2026). Violência epistêmica algorítmica: fundamentos teóricos e éticos de uma IA decolonial. *Revista Tempo Psicanalítico*, 58, e-952. <https://doi.org/10.71101/rtp.58.952>
- [4] Chen, J., Xiao, S., Zhang, P., Luo, K., Lian, D., & Liu, Z. (2025). M3-Embedding: Multilinguality, multi-functionality, multi-granularity text embeddings through self-knowledge distillation. arXiv:2402.03216.
- [5] Kay, J., Kasirzadeh, A., & Mohamed, S. (2024). Epistemic injustice in generative AI. arXiv:2408.11441.
- [6] Lewis, P., et al. (2020). Retrieval-augmented generation for knowledge-intensive NLP tasks. *Advances in Neural Information Processing Systems*, 33, 9459–9474.
- [7] Mollema, W. J. T. (2025). A taxonomy of epistemic injustice in the context of AI and the case for generative hermeneutical erasure. *AI and Ethics*, 5, 5535–5555.
- [8] Noble, S. U. (2018). *Algorithms of oppression: How search engines reinforce racism*. NYU Press.

- [9] Santos, B. S. (2014). *Epistemologies of the South: Justice against epistemicide*. Paradigm Publishers.
- [10] Spivak, G. C. (2010). Can the subaltern speak? In R. C. Morris (Ed.), *Can the subaltern speak? Reflections on the history of an idea* (pp. 237–291). Columbia University Press. (Original work published 1988)